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MECHANICAL VIBRATIONS L3

FORCED VIBRATION OF 1 DOF

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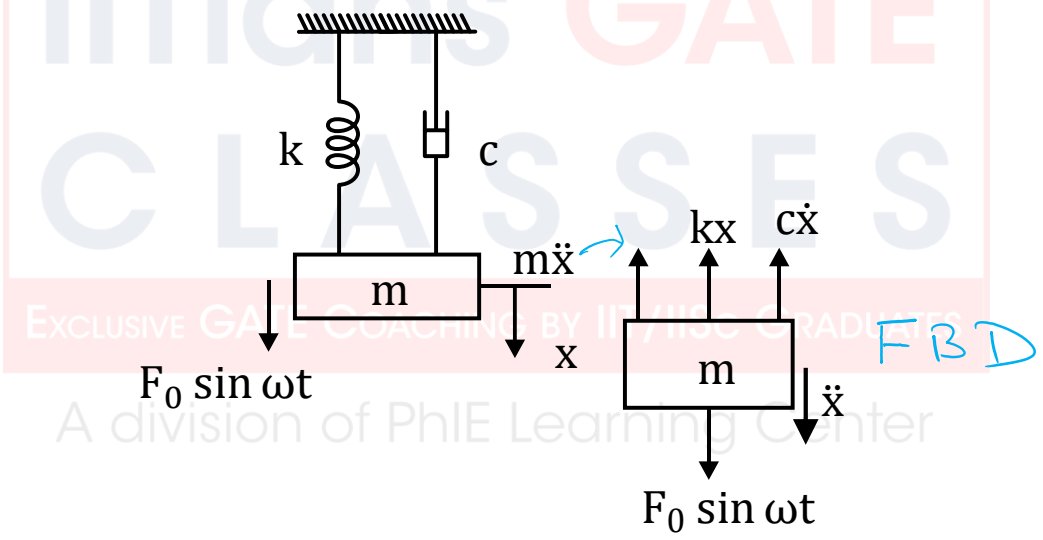
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DAMPED FORCED VIBRATION OF 1 DOF

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Consider a spring mass system having viscous damping, excited by a sinusoidal forcing function $F_0 \sin \omega t$ as shown in fig.



$$m\ddot{x} + c\dot{x} + kx = F_0 \sin \omega t$$

EQUATION OF MOTION

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$$m\ddot{x} + c\dot{x} + kx = F_0 \sin \omega t$$

This is a linear, non-homogeneous, second order differential equation.

The solution of this equation consists of two parts – complementary and particular integral.

$$x = x_c + x_p$$

The complementary function(x_c) is obtained from

$$m\ddot{x} + c\dot{x} + kx = 0$$

$$x_c = A_2 e^{-\zeta \omega_n t} \sin\left[\left(\sqrt{1-\zeta^2}\right)\omega_n t + \phi_2\right]$$

Considering under damped system due oscillation by excitation force

EQUATION OF MOTION

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If the particular solution is x_p,

then $x_p = X\sin(\omega t - \phi)$

1st 2nd

$$X = \frac{X_{st}}{\sqrt{\left[1-\left(\frac{\omega}{\omega_n}\right)^2\right]^2 + \left[2\zeta\frac{\omega}{\omega_n}\right]^2}}$$

$$\phi = \tan^{-1}\left[\frac{2\zeta\frac{\omega}{\omega_n}}{1-\left(\frac{\omega}{\omega_n}\right)^2}\right]$$

$x = x_c + x_p$

not natural frequency

$m\ddot{x} + kx + c\dot{x} = F_0\sin\omega t$

$[mD^2 + k + cD]x_p = F_0\sin\omega t$

$x_p = \frac{F_0\sin\omega t}{[mD^2 + k + cD]}$

Excitation frequency.

transient ← which is changing with time

Amplitude

↓ viceversa

steady Amplitude

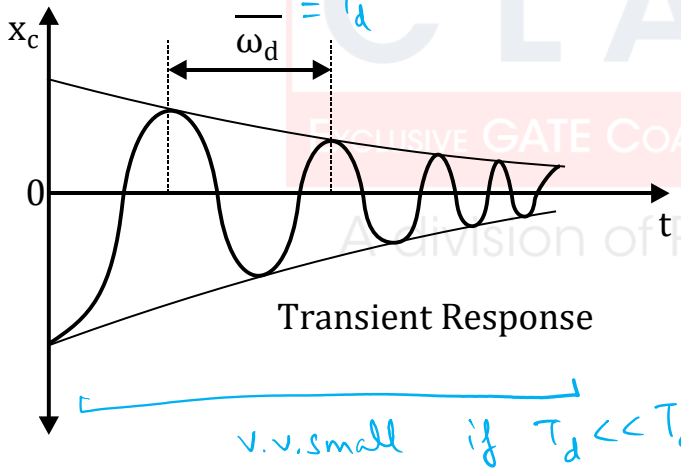
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EQUATION OF MOTION

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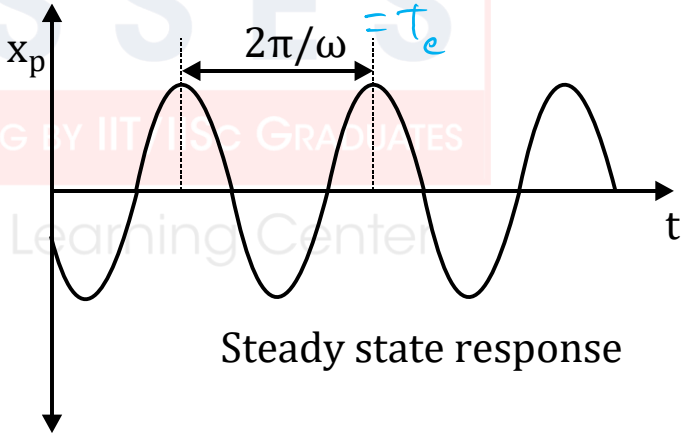
$$x = x_c + x_p$$

$$= A_2 e^{-\zeta\omega_n t} \sin\left[\left(\sqrt{1-\zeta^2}\right)\omega_n t + \phi_2\right] + \frac{X_{st}}{\sqrt{\left[1-\left(\frac{\omega}{\omega_n}\right)^2\right]^2 + \left[2\zeta\frac{\omega}{\omega_n}\right]^2}} \sin(\omega t - \phi)$$



Transient Response

v.v.small if $T_d << T_e$



Steady state response

$$X_{steady\ state} = \frac{X_{static}}{\sqrt{\left[1-\left(\frac{\omega}{\omega_n}\right)^2\right]^2 + \left(2\xi_1 \frac{\omega}{\omega_n}\right)^2}}$$

$$X_{static} = \frac{F_0}{k}$$



Excitation frequency $\frac{\omega}{\omega_n} = r$ ← frequency ratio

Natural frequency of system

Static: $F = F_0 \Rightarrow F_0 = kX_{static} \Rightarrow X_{static} = \frac{F_0}{k}$

Dynamic: $F = F_0 \sin \omega t$

$$X_{steady\ state} = \frac{X_s}{\sqrt{\left[1-r^2\right]^2 + \left[2\xi r\right]^2}}$$

where $X_s = F_0/k$, $r = \frac{\omega}{\omega_n}$

$$\phi = \tan^{-1}\left(\frac{2\xi r}{1-r^2}\right)$$

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MAGNIFICATION FACTOR

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The ratio of the steady state amplitude to the zero frequency (static) deflection

i.e., $\frac{X}{X_{st}}$ is defined as the magnification factor and is denoted by M.F.

$$M.F. = \frac{X}{X_{st}} = \frac{1}{\sqrt{\left[1-\left(\frac{\omega}{\omega_n}\right)^2\right]^2 + \left[2\zeta\frac{\omega}{\omega_n}\right]^2}} = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}$$

$$M.F. = \frac{X}{X_{st}} = \frac{1}{\sqrt{[1-(r)^2]^2 + [2\zeta r]^2}}$$

$x = X\sin(\omega t + \phi)$ ← Equation form independent of ranges of ζ

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MAGNIFICATION FACTOR

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$$\text{frequency ration } (r) = \frac{\omega}{\omega_n}$$

$$\text{Static deflection } (X_{st}) = \frac{F_0}{k}$$

Undamped system ($\xi = 0$) :

$$M \cdot F = \frac{1}{[1 - r^2]} \qquad \text{when, } r = 1, M.F = \text{infinity (resonance)}$$

Underdamped & Overdamped system ($0 < \xi$) :

$$M.F = \frac{1}{\sqrt{(1 - r^2)^2 + (2\xi r)^2}}$$

MAGNIFICATION FACTOR

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Maximum MF Conditions

$$\frac{dM}{dr} = 0 \Rightarrow \boxed{r = \sqrt{1 - 2\xi^2}}$$

$0 < \xi \leq 1/\sqrt{2}$

$r = \sqrt{+\text{value}}$

Real values

$1/\sqrt{2} < \xi < 1$

$r = \sqrt{-\text{value}}$

Imaginary values

No maximum (Monotonically decreases)

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MAGNIFICATION FACTOR

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At maximum MF conditions: (or peak amplitude conditions)

$$r = \sqrt{1 - 2\xi^2}$$

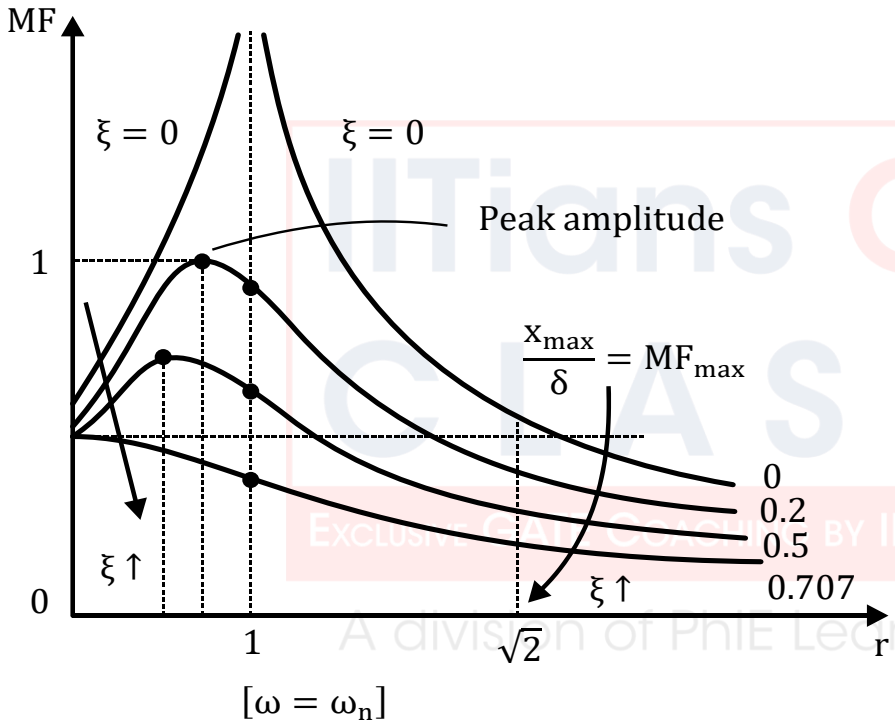
$$MF = \frac{1}{\left[[1 - (2 - 2\xi^2)]^2 + 4\xi^2(1 - 2\xi^2)\right]^{1/2}} = \frac{1}{2\xi\sqrt{1 - \xi^2}}$$

$$\boxed{MF_{max} = \left[\frac{x_{max}}{X_{st}}\right] = \frac{1}{2\xi\sqrt{1 - \xi^2}}}$$

occuring at $r = \sqrt{1-2\xi^2}$ only for $0 < \xi < \frac{1}{\sqrt{2}}$

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MAGNIFICATION FACTOR

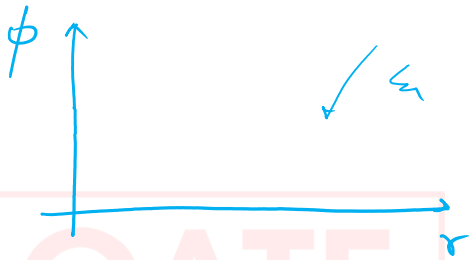


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PHASE ANGLE OBSERVATIONS

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$$\phi = \tan^{-1}\left\{\frac{2\zeta\frac{\omega}{\omega_n}}{1-\left(\frac{\omega}{\omega_n}\right)^2}\right\} = \tan^{-1}\left(\frac{2\zeta r}{1-r^2}\right)$$



1. For an undamped system ($\zeta = 0$), shows that the phase angle is 0 for $0 < r < 1$ and $180°$ for $r > 1$.

   This implies that the excitation and response are in phase for $0 < r < 1$ and out of phase for $r > 1$ when $\zeta = 0$.

2. For $\zeta > 0$ and $0 < r < 1$, the phase angle is given by $0 < \phi < 90°$, implying that the response lags the excitation.

$1-r^2 > 0$

$\uparrow \phi \propto \zeta \uparrow$

$F = F_0 \sin(\omega t)$

$x = X \sin(\omega t + \phi)$

PHASE ANGLE OBSERVATIONS

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3. For $\zeta > 0$ and $r = 1$, the phase angle is given by $90° < \phi < 180°$, implying that the response lead the excitation.
4. For $\zeta > 0$ and $r = 1$, the phase angle is given by $\phi = 90°$, implying that the phase difference between the excitation and the response if $90°$.
5. For $\zeta > 0$ and large values of r, the phase angle approaches $180°$, implying that the response and the excitation are out or phase.

PHASE ANGLE OBSERVATIONS

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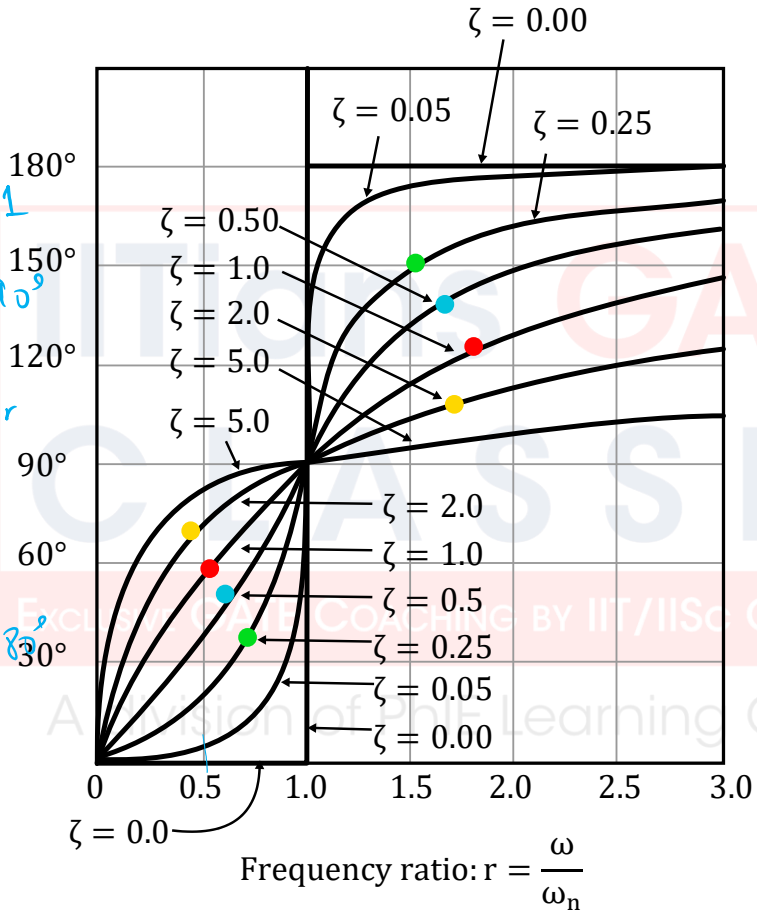
Conclusion:-

1) For a particular value of $0 < r < 1$

$\Rightarrow \phi \uparrow$ & $0 < \phi < 90°$

2) For a particular value of $1 < r$

$\Rightarrow \phi \downarrow$ & $90° < \phi < 180°$



$\phi = \tan^{-1}\left(\frac{2\xi r}{1-r^2}\right)$

$r = 1$

$\phi = \tan^{-1}\left(\frac{2\xi}{0}\right)$

$= \tan^{-1}(\infty)$

$= \frac{\pi}{2}$

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QUALITY FACTOR

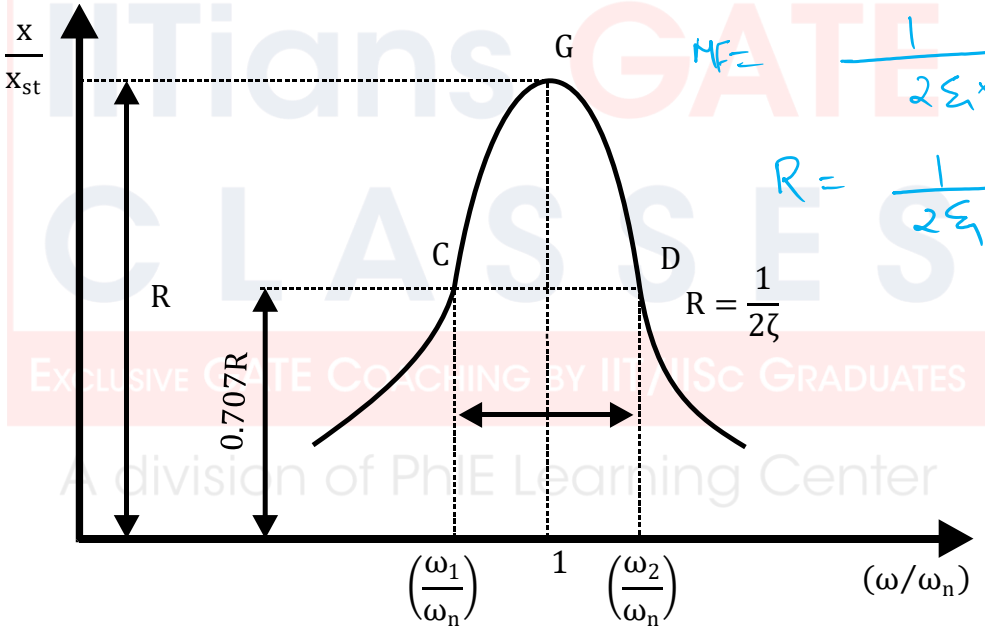
The magnification factor at resonance is known as quality factor and is denoted by R.

$$R = \frac{1}{2\zeta}$$

$$MF = \frac{1}{\sqrt{(1-r^2)^2 + (2\xi r)^2}}$$

$$MF = \frac{1}{2\xi \times 1}$$

$$R = \frac{1}{2\xi} = \frac{X}{x_{st}}$$



C and D are the half power points.

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HALF POWER POINTS

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Half-power points are the two frequencies on either side of the resonant frequency at which the **power of the vibrating system becomes half of its maximum value**

Assume ω_1 and ω_2 are frequency corresponding to points C and D respectively.

$$\frac{\omega_2 - \omega_1}{\omega_n} = 2\zeta = Band\ width = \frac{1}{R}$$

$$\rightarrow Energy\ absorbed\ at\ point\ G\ is\ \frac{1}{2}m\omega^2R^2$$

$E = \frac{1}{2} m \omega^2 A^2$

$E_{max} = \frac{1}{2} m \omega^2 R^2$

$$\rightarrow Energy\ absorbed\ at\ point\ C\ \&\ D\ is\ \frac{1}{2}m\omega^2\left(\frac{R}{\sqrt{2}}\right)^2 = \frac{1}{2}\left(\frac{1}{2}m\omega^2R^2\right)$$

$$E_{max} = \frac{1}{2} m\omega^2 R^2$$

$$E = \frac{E_{max}}{2} \qquad \frac{\frac{1}{2} m\omega^2 R^2}{2}$$

$$E = \frac{1}{2} m\omega^2 \left(\frac{R}{\sqrt{2}}\right)^2$$

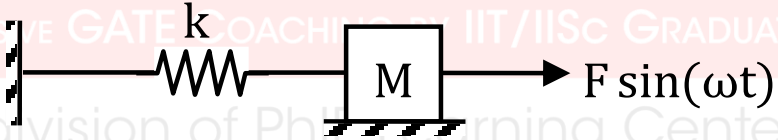
$$E = \frac{1}{2} m\omega^2 \left(0.707 R\right)^2$$

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PROBLEM 1

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A single degree of freedom spring-mass system is subjected to a harmonic force of constant amplitude. For an excitation frequency of $\sqrt{3k/m}$, the ratio of the amplitude of steady state response to the static deflection of the spring is _____



Given :- $\omega = \sqrt{\frac{3k}{m}}$, $\omega_n = \sqrt{\frac{k}{m}}$, $c = 0 \Rightarrow \xi = 0$

$$MF = \frac{1}{\sqrt{(1-r^2)^2 + (2\xi r)^2}}$$

$$MF = \frac{1}{\sqrt{(1-r^2)^2 + 0}}$$

$\therefore r = \frac{\omega}{\omega_n} = \sqrt{3}$

$\Rightarrow MF = \frac{1}{|(1-3)^2|}$

$= \frac{1}{2}$

$= 0.5$

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Given:- $W =$

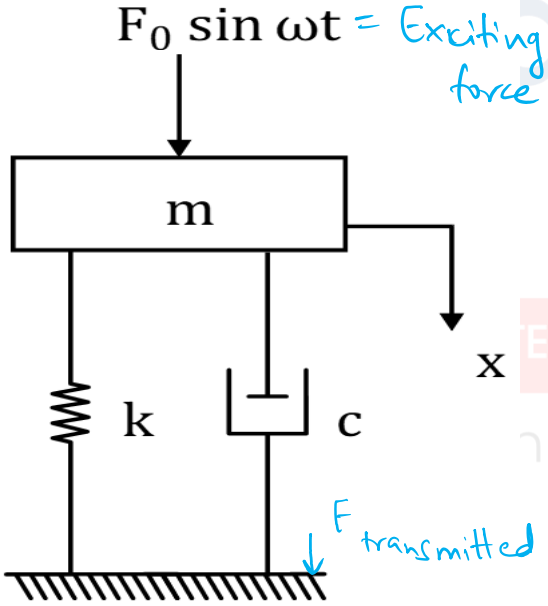
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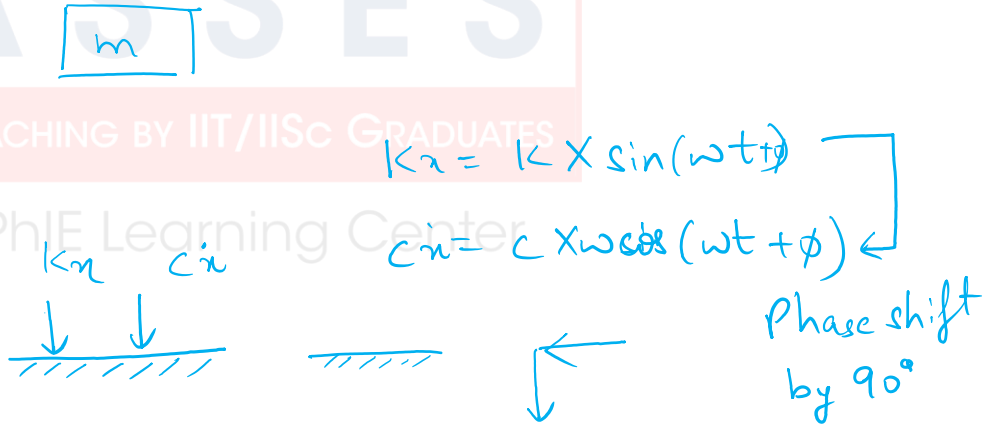
FORCE TRANSMISSIBILITY

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Force transmissibility is the **ratio of the amplitude of force transmitted to the support** to the **amplitude of the applied (exciting) force** in a vibrating system.



$$\text{Force Transmissibility } (T_F) = \frac{\text{Transmitted force}}{\text{Exciting force}}$$



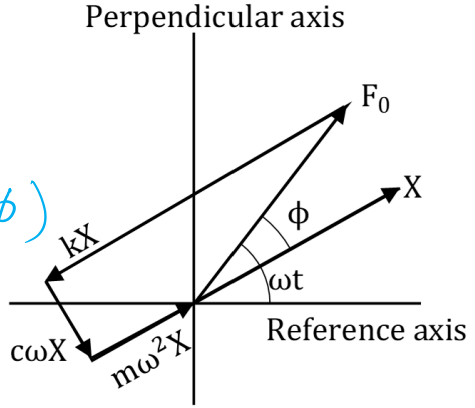
FORCE TRANSMISSIBILITY

$$F_{tr} = \sqrt{(kX)^2 + (c\omega X)^2} = X\sqrt{k^2 + (c\omega)^2}$$

$$F_{tr} = \frac{F_o\sqrt{k^2 + (c\omega)^2}}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}$$

$$T_F = \frac{F_{tr}}{F_o} = \frac{\sqrt{1 + \left(2\zeta\frac{\omega}{\omega_n}\right)^2}}{\sqrt{\left[1 - \left(\frac{\omega}{\omega_n}\right)^2\right]^2 + \left[2\zeta\frac{\omega}{\omega_n}\right]^2}}$$

T_F being the transmissibility



$F = F_0 \sin(\omega t)$

$x = X \sin(\omega t \mp \phi)$

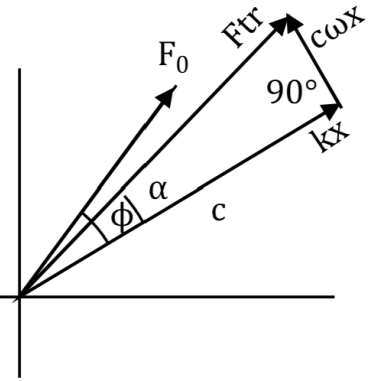
kx

$c\omega X$ ← 90° phase shift

$$X = \frac{X_{st}}{\sqrt{(1-r^2)^2 + (2\xi r)^2}}$$

$$F_{tr} = \frac{F_0\sqrt{1 + (2\xi r)^2}}{\sqrt{(1-r^2)^2 + (2\xi r)^2}}$$

$$T_F = \frac{F_{tr}}{F_0}$$



$$F_{tr} = X\sqrt{k^2+(c\omega)^2}$$

$$F_{tr} = \frac{F_0\sqrt{k^2+(c\omega)^2}}{k\sqrt{(1-r^2)^2+(2\xi r)^2}}$$

$$= F_0\frac{\sqrt{1+\left(\frac{c\omega}{k}\right)^2}}{\sqrt{(1-r^2)^2+(2\xi r)^2}}$$

$$\frac{c\omega}{k} = \frac{c}{k}\omega$$

$$= \frac{\xi\, 2\sqrt{mk}\,\omega}{k}$$

$$\xi = \frac{c}{2\sqrt{mk}}$$

$$F_{tr} = \frac{F_0\sqrt{1+(2\xi r)^2}}{\sqrt{(1-r^2)^2+(2\xi r)^2}}$$

$$= \xi\, 2\sqrt{\frac{m}{k}}\,\omega$$

$$\frac{c\omega}{k} = 2\xi\frac{\omega}{\omega_n}$$

$$\boxed{\frac{c\omega}{k} = 2\xi r}$$

$$\Rightarrow T_F = \frac{F_{tr}}{F_0} = \frac{\sqrt{1+(2\xi r)^2}}{\sqrt{(1-r^2)^2+(2\xi r)^2}}$$

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FORCED VIBRATION DUE TO EXCITATION OF SUPPORT

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Forced vibrations due to the excitation of the support

$$m\ddot{x} = -c(\dot{x} - \dot{y}) - k(x - y)$$

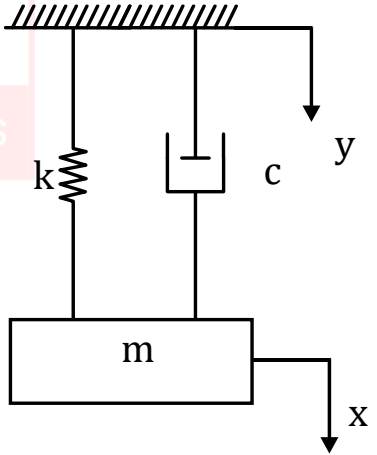
$$m\ddot{x} + c\dot{x} + kx = ky + c\dot{y} = A\sin(\omega t - \alpha)$$

This shows that giving excitation to the base is equivalent to applying a harmonic force of magnitude A to the mass.

$$A = Y\sqrt{k^2 + (c\omega)^2}$$

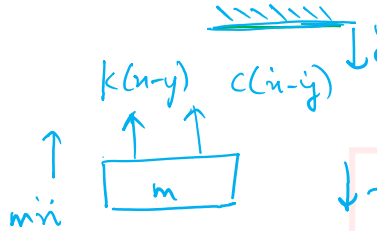
$$\alpha = tan^{-1}\left(\frac{-c\omega}{k}\right)$$

$y = Y\sin\omega t$



FORCED VIBRATION DUE TO EXCITATION OF SUPPORT

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$$k(x-y) + c(\dot{x}-\dot{y}) + m\ddot{x} = 0$$

$$kx + c\dot{x} + m\ddot{x} = ky + c\dot{y}$$

$$= kY\sin\omega t + c\omega Y\cos\omega t$$

$$= \sqrt{(kY)^2 + (c\omega Y)^2}\left[\frac{kY}{A}\sin\omega t + \frac{c\omega Y}{A}\cos\omega t\right]$$

$\frac{kY}{A} = \cos\alpha$, $\frac{c\omega Y}{A} = \sin\alpha$

$$kx + c\dot{x} + m\ddot{x} = Y\sqrt{k^2 + (c\omega)^2}\sin(\omega t + \alpha)$$

$$F_0\sin(\omega t) = kx + c\dot{x} + m\ddot{x} = A\sin(\omega t + \alpha) \rightarrow x_p = ?$$

where $A = Y\sqrt{k^2 + (c\omega)^2}$

$x_p = X\sin(\omega t + \phi)$

$$X = \frac{F_0}{k\sqrt{(1-r^2)^2 + (2\xi r)^2}}$$

$$\tan\phi = 2\xi r/(1-r^2)$$

$$\tan\alpha = \frac{\frac{c\omega Y}{A}}{\frac{kY}{A}} = \frac{c\omega}{k}$$

$$\Rightarrow \alpha = \tan^{-1}\left(\frac{c\omega}{k}\right)$$

FORCED VIBRATION DUE TO EXCITATION OF SUPPORT

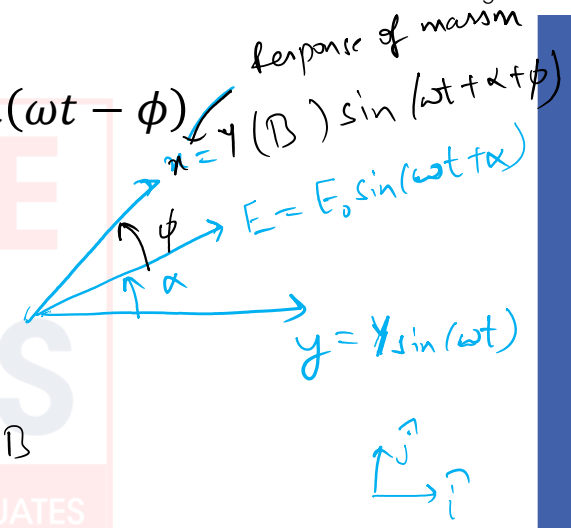
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$$x_p(t) = \frac{Y\sqrt{k^2 + (c\omega)^2}}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}} \sin(\omega t - \phi_1 - \alpha) = X \sin(\omega t - \phi)$$

$$\phi_1 = \tan^{-1}\left(\frac{c\omega}{k - m\omega^2}\right)$$

$$\frac{X}{Y} = \left[\frac{k^2 + (c\omega)^2}{(k - m\omega^2)^2 + (c\omega)^2}\right]^{1/2} = \left[\frac{1 + (2\zeta r)^2}{(1 - r)^2 + (2\zeta r)^2}\right]^{1/2}$$

$$\phi = \tan^{-1}\left[\frac{mc\omega^3}{k(k - m\omega^2) + (\omega c)^2}\right] = \tan^{-1}\left[\frac{2\zeta r^3}{1 + (4\zeta^2 - 1)r^2}\right]$$



FORCED VIBRATION DUE TO EXCITATION OF SUPPORT

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$$X = Y\sqrt{\frac{1+\left(\frac{c\omega}{k}\right)^2}{(1-r^2)^2+(2\xi r)^2}}$$

$$\frac{c\omega}{k} = 2\xi r$$

$$x_{st} = \frac{F_0}{k\sqrt{(1-r^2)^2+(2\xi r)^2}}$$

$$F_0 = Y\sqrt{k^2+(c\omega)^2}$$

$$x_{st} = \frac{Y\sqrt{k^2+(c\omega)^2}}{k\sqrt{(1-r^2)^2+(2\xi r)^2}}$$

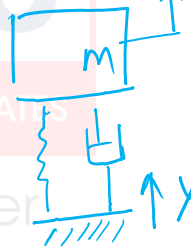
$$\frac{X}{x_{st}} = Y\sqrt{\frac{1+(2\xi r)^2}{(1-r^2)^2+(2\xi r)^2}}$$

$$T_d = \frac{X}{Y} = \cdots$$

1st case



2nd case



F_t

$$T_F = \frac{F_t}{F_0} \leftarrow \text{Effect}$$

$$T_d = \frac{X}{Y} \leftarrow \text{Effect}$$

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$$T_d = \frac{\sqrt{1+(2\xi r)^2}}{\sqrt{(1-r^2)^2+(2\xi r)^2}}$$

$\frac{X}{Y}$ → **displacement transmissibility (T_d)**

r, ξ, T_d ← output

1) $r = 0 \Rightarrow T_d = \frac{1}{\sqrt{1+0}} = 1$ for any ξ

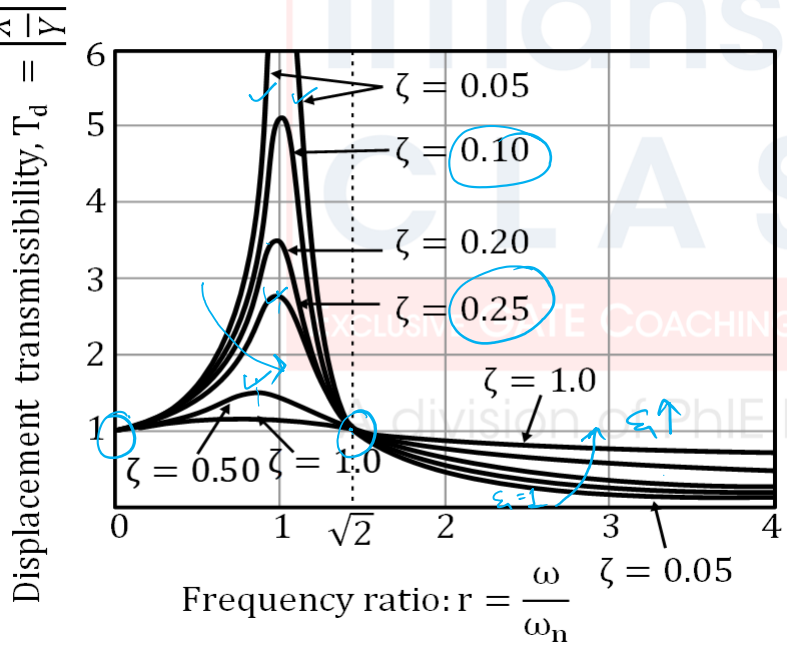
2) $\xi = 0 \Rightarrow T_d = \frac{1}{|1-r^2|}$

$r < 1 \Rightarrow T_d \propto \left(\frac{1}{<1}\right) > 1$

$r = 1 \Rightarrow T_d \to \infty$

$\sqrt{2} > r > 1 \Rightarrow T_d \propto \left(\frac{1}{<1}\right) > 1$ $\quad T_d|_{r=\sqrt{2}} = \frac{1}{1}$

$r > \sqrt{2} \Rightarrow T_d \propto \left(\frac{1}{>1}\right) < 1$



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$$T_d = \frac{\sqrt{1+(2\xi r)^2}}{\sqrt{(1-r^2)^2+(2\xi r)^2}}$$

3) $r=\sqrt{2} \Rightarrow T_d = \dfrac{\sqrt{1+(2\xi\sqrt{2})^2}}{\sqrt{1+(2\xi\sqrt{2})^2}}$

$\therefore \forall\, \xi,\ r=\sqrt{2} \Rightarrow T_d = 1$

4) $r=1$ (Resonance condn)

$\Rightarrow T_d = \dfrac{\sqrt{1+(2\xi r)^2}}{\sqrt{0+(2\xi r)^2}} = \sqrt{\dfrac{1+(2\xi)^2}{(2\xi)^2}} = \sqrt{\dfrac{1}{(2\xi)^2}+1}$

$\Rightarrow T_d \propto \dfrac{1}{\xi\uparrow}$

As Damping is ↑ ⇒ Effect of excitation is decreasing

$\Rightarrow \xi \to \infty \Rightarrow T_d \to 1$

5) $r>\sqrt{2}\ \forall\ \xi$, $r=3$

$\Rightarrow T_d = \dfrac{\sqrt{1+(6\xi)^2}}{\sqrt{64+(6\xi)^2}} = \dfrac{N^r}{D^r} \Rightarrow D^r > N^r \Rightarrow T_d < 1$

$\Rightarrow \xi=1 \Rightarrow T_{d,1} = \dfrac{\sqrt{7}}{\sqrt{64+36}} = \dfrac{}{D_1^r} \simeq 0$ $\quad \xi=2 \Rightarrow T_{d,2} = \dfrac{\sqrt{145}}{\sqrt{64+144}} = \dfrac{}{D_2^r}$ $\Rightarrow D_2^r > D_1^r \Rightarrow T_{d,2} < T_{d,1}$

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- The value of T_d is unity at $r = 0$ and close to unity for small values of r.
- For an undamped system ($\zeta = 0$), $T_d \to \infty$ at resonance ($r = 1$).
- The value of T_d is less than unity ($T_d < 1$) for values of $r > \sqrt{2}$ (for any amount of damping ζ).
- The value of T_d is unity for all values of ζ at $r = \sqrt{2}$.
- For $r < \sqrt{2}$, smaller damping ratios lead to larger values of T_d. On the other hand for $r > \sqrt{2}$, smaller damping ratios lead to smaller values of T_d.

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- The displacement transmissibility, T_d, attains a maximum for $0 < \zeta < 1$ at the frequency ratio $r = r_m < 1$ given by

$$r_m = \frac{1}{2\zeta}\left[\sqrt{1 + 8\zeta^2} - 1\right]^{1/2}$$

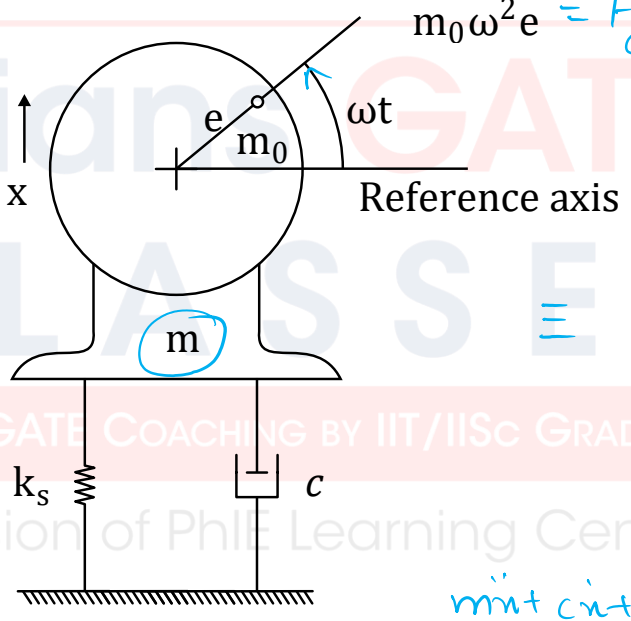
$0 < r < \sqrt{2}$

$r_m \propto \frac{1}{\sqrt{\zeta}}$

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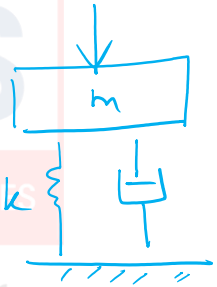
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Rotating unbalance

F_0
$F = \overbrace{me\omega^2}\sin(\omega t)$



$m\ddot{x} + c\dot{x} + kx = F_0\sin(\omega t)$

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$$m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = m_0 e\omega^2 \sin\omega t$$

$$X = \frac{\frac{m_0 e\omega^2}{k}}{\sqrt{\left(1 - \frac{m\omega^2}{k}\right)^2 + \left(\frac{c\omega}{k}\right)^2}}$$

$$X = \frac{F_0}{k\sqrt{(1-r^2)^2 + (2\xi r)^2}}$$

$$X = \frac{m_0 e\omega^2}{k\sqrt{(1-r^2)^2 + (2\xi r)^2}}$$

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$$\frac{X}{\frac{m_0 e\omega^2}{k}} = -\frac{\left(\frac{\omega}{\omega_n}\right)^2}{\sqrt{\left[1-\left(\frac{\omega}{\omega_n}\right)^2\right]^2+\left[2\zeta\frac{\omega}{\omega_n}\right]^2}}$$

$$\phi = \tan^{-1}\left[\frac{2\zeta\frac{\omega}{\omega_n}}{1-\left(\frac{\omega}{\omega_n}\right)^2}\right]$$

At resonance when $\frac{\omega}{\omega_n} = 1$, we have

$$MF = \frac{X}{\frac{m_0 e\omega^2}{k}} = \frac{1}{2\zeta}$$

$$X = \frac{m_0 e\omega^2}{k\sqrt{(1-r^2)^2+(2\xi r)^2}}$$

$$X = \frac{m_0 e\omega^2}{k(2\xi)}$$

$$MF = \frac{1}{2\xi} = \frac{XK}{m_0 e\omega^2}$$

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PROBLEM 2

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A single-degree-freedom spring-mass system is subjected to a sinusoidal force of 10 N amplitude and frequency ω along the axis of the spring. The stiffness of the spring is 150 N/m, damping factor is 0.2 and the undamped natural frequency is 10ω. At steady state, the amplitude of vibration (in m) is approximately

(A) 0.05

(B) 0.07

(C) 0.70

(D) 0.90

Answer: B

Given :- $F = 10\sin(\omega t)$, $\xi = 0.2$, $k = 150\ N/m$, $\omega_n = 10\omega$

$X = ?$

$$r = \frac{\omega}{\omega_n} = \frac{\omega}{10\omega} = 0.1$$

$$X = \frac{F_0}{k\sqrt{(1-r^2)^2 + (2\xi r)^2}} = \frac{10}{150\sqrt{(1-0.01)^2 + (2\times 0.2\times 0.1)^2}}$$

$$= \frac{1}{15\sqrt{0.98 + 0.0016}}$$

$$= \frac{1}{15\sqrt{0.9816}}$$

$$\approx 0.074$$

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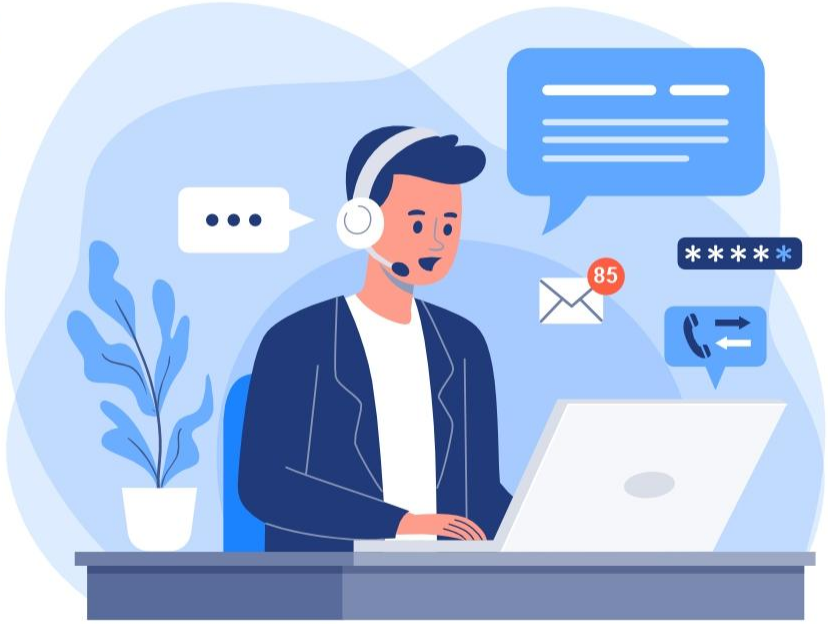
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